

Similarity-based inferences predict cross-talker generalization during human speech perception

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Abstract

Human speech perception is now understood to be highly adaptive. Unfamiliar regional or second language ‘accents’ might map acoustics to sound categories and word meanings in ways that deviate strongly from listeners’ expectations. But the initial processing difficulties associated with unfamiliar pronunciations patterns tend to ameliorate within minutes of exposure. Perhaps even more remarkably, listeners seem to be able to generalize this adaptation to novel talkers of the same accent and sometimes even to talkers of other accents. We use modern deep learning models to examine the mechanisms that afford such rapid generalization.

Keywords: speech perception, adaptation, generalization, naturalistic speech

Introduction

Human speech perception relies on effective generalization of past experience to unfamiliar inputs. Pronunciation of the same word can vary substantially based on talkers’

vocal tract physiology, social identity, and language background. Yet, to the surprise of pioneer speech scientists, this “lack of invariance” [1] rarely is disruptive for human speech perception [2]. When listeners encounter unfamiliar speech patterns—such as unfamiliar second language (L2) accents—much of the initial decrease in accurate speech recognition dissipates within 1-2 minutes of exposure [3–5]. This rapid adaptivity is made possible by effective generalization from previously experienced speech inputs to the new, unfamiliar input [6–9].

The perceptual mechanisms that enable such rapid generalization remain poorly understood (for review [10]). We operationalize two influential hypotheses, and evaluate their ability to quantitatively explain human generalization behavior during adaptive speech perception. The first hypothesis holds that *high variability exposure* (HVE) is critical for successful learning and generalization [5]. Under the HVE hypothesis, listeners require sufficiently variable exposure to distinguish talker- from accent-level variability and thereby form the talker-independent accent representations necessary for cross-talker generalization. Support for this hypothesis comes from findings that successful generalization requires exposure to multiple talkers [5, 9]. The HVE hypothesis has since shaped research on speech perception and beyond, including fields ranging from pedagogy to rehabilitation therapy [?].

Recent research, however, suggests a need to revisit the HVE hypothesis, finding cross-talker generalization even after short exposure to a single talker [6, 7, 11]. Similarity-based inference (SBI), in contrast, proposes that listeners generalize to the extent that exposure and test speech share similar mappings from perceptual representations to speech categories [6, 11]. Under SBI, high exposure variability therefore is *neither necessary, nor sufficient* for generalization. If speech inputs during test are similar to those experienced during exposure, successful generalization should occur even after short exposure to a single talker; conversely, high variability exposure should benefit listeners if, and only if, high variability exposure includes inputs that have relevant similarities to test inputs.

Despite extensive research, fundamental questions about the processes underlying generalization during adaptive speech perception remain unresolved [12]. Competing hypotheses about cross-talker generalization, including HVE and SBI, are verbal theories, making it difficult to systematically evaluate their explanatory power. Fully-specific computational models are needed to quantify whether—and *how well*—these hypotheses can explain generalization during everyday speech perception. Existing computational models of adaptive speech [13–15] provide important theoretical insights but do not readily scale to the type of natural, acoustically heterogeneous, speech that listeners encounter outside the laboratory [16]. To overcome these limitations, we develop computational implementations of HVE and SBI within a common modeling framework that combines a general cognitive theory of speech perception (exemplar theory [13]) with high-dimensional speech representations learned by modern deep neural networks (DNNs) through only self-supervision—i.e., without external feedback that would not be available to human learners. This framework enables direct quantitative comparison of competing theories across multiple datasets and allows us to identify what speech properties explain cross-talker generalization in human speech perception.

Results

We used this DNN-based approach (Fig 1) to evaluate the HVE and SBI against three published datasets on listeners’ ability to generalize within and across talkers and second language (L2) accents (total $N = 675$ listeners). The combined data contain exposure to a broad range of L2-English accents by L1-talkers of different languages (Fig 1c). The sound structures of these L1s deviate in different ways from English (Fig 1d), leading to distinctive L2 English accent properties. For example, Spanish and Mandarin have substantially simpler vowel inventories than English, which can lead to L2-English vowel realization that deviate from L1-English pronunciations (e.g., “peach” pronounced as “pitch”). Korean has substantially simpler syllable structure than English, and L1-Korean learners of English sometimes simplify English syllable structure by omitting and/or inserting additional segments (e.g., “street” pronounced as “seu-teu-rit” or “bench” pronounced as “bench-ee”). Any of these unexpected accent- or talker-specific speech properties can initially cause processing difficulty for L1-English listeners [3, 4, 6].

figures/computational-approach.png

Fig. 1: Computational approach to obtain latent speech representations. **a**, speech recordings are transformed into a continuous latent feature representation. Speech recordings are projected into a layer of a DNN pre-trained with self-supervised statistical learning on large amounts of English speech (HuBERT [17]). A non-linear transformation reduces the dimensionality of the feature representation, while maintaining critical information (t -SNE [18]). This transforms recordings of duration T into $T \times 3$ matrices $\in \mathbb{N}^+ \times \mathbb{R}$, representing the speech signal over each time interval $t \in \{1, \dots, T\}$ as 3D feature vector. **b**, instances of segments, words, and sentences form ‘hyper-noodles’ in this 3D space, shown for segments XXX, XXX, and XXX in the word “XXX” spoken by a male, first-language (L1) speaker of North American English. **c**, first language (L1) backgrounds represented in the data, colored by language branch. All experiments investigate how L1-English listeners perceive L2-English by talkers of different L1-backgrounds (for details, see Table 1). **d**, L1’s sound structure similarity to English (segmental and supra-segmental phonology, phonotactics) based on PHOIBL 2.0 [19].

The DNN-based approach successfully recovers well-documented differences between the speech of native/first-language (L1) talkers and L2 talkers (Fig 2a-b; see also [20]). These pronunciation differences predict how intelligible a particular set of L2 speech recordings is for L1 listeners (Fig 2c; see also [21]) and the types of recognition errors L1 listeners experience when listening to that L2 speech (Fig 2d). This suggests that the latent feature representations learned by the DNN resemble those acquired by human learners sufficiently well for the current purpose (see discussion in [22]).

figures/feasibility.png

Fig. 2: Latent representations learned through self-supervised learning capture important properties of the production and perception of second language (L2) speech. **a**, word-level pronunciation similarity between 36 talkers from the AN19 data, depending on the talker’s L1. **b**, magnitude of L2-to-L1 deviation from L1-English pronunciations by different L2 talkers in AN19 data for selected segment types (phonemes). Vertical lines show mean across talker of the same L1 background, shaded intervals show 95% CI. **c**, L2-English pronunciation similarity to L1-English predicts intelligibility by L1-English listeners prior to informative exposure in AN19 (Korean- and Spanish-accented tests in control condition) and X21 data (Mandarin-accented test in control). **d**, segment-level perception errors in AN19 (control) as a function of L2-to-L1 deviation (panel b) and lexical constraint (number of minimal pair word neighbors only distinguished by the segment).

figures/variability-approach.png

Fig. 3: Quantifying speech variability. **a**, inferred latent representations (Tr layer 24) for all exposure speech recordings in two of the exposure conditions in the X21 data: L2-English speech from (top) five L1-Mandarin talkers, each producing the same 16 sentences vs. (bottom) from a single L1-Mandarin talker. **b**, illustrating how between- and within-segment variability was calculated from these latent presentations. **c**, distribution of between- and within-segment variability across the four exposure conditions in the X21 data. Each point summarizes a particular stimulus list, as some exposure conditions had multiple instantiations that varied in the specific speech stimuli listeners were exposed to. **d** ...

figures/similarity-approach.png

Fig. 4: Quantifying exposure-test similarity in the mappings from acoustics to speech categories. **a**, ...

The SBI hypothesis holds that

- Explanation beyond original exposure conditions. compare exp-test-sim against original condition. quantify within and between-condition variability captured.

figures/layer-wise.png

Fig. 5: Latent representations in later DNN layers capture more relevant information about human behavior **a**, proportion of explainable variance in AN19 explain by HVE and SBI across layers of the DNN. For comparison, baseline performance for low-level representations of acoustic (Mel-Fourier Cepstral Coefficients, MFCCs) and auditory information (spectro-temporal receptive fields, STRFs) is provided on the left. Horizontal dashed line indicates Bonferroni-corrected threshold for significance. **b**, same for X21. **c**, same for B23. **d**, correlation matrix summarizing how exposure-test similarity across the different layers for all three datasets.

Discussion

Our findings demonstrate that it is possible to use modern DNN-based ASR systems to scale theories of speech perception to natural speech (see also [20, 21]). This does not mean that the representations learned by modern ASR systems are identical to those learned by human listeners. As long as the DNN architectures underlying modern ASR systems learned latent representations that induce exposure-test (dis)similarities that resemble those experienced by listeners, this can explain our results.

- Discuss: Chernyak and Kim work.
- Caveat: Learning is ‘like’ human learning but also quite different (large amounts of memory and revisiting the same text).
- rodriguez2024: qualitative comparison of whether model struggles with non-native perception in ways listeners do. Finds that model is sensitive to different features.

Conclusion

Methods

Data

We processed three published datasets of L1-English listeners’ adaptation to L2-English speech [6–8]. All three datasets were collected in exposure-test experiments (total $N = 675$ listeners). Each experiment randomly assigned listeners to one of multiple *exposure conditions* that differed in the type of speech listeners heard—e.g., L1 or L2 speech by one or more talkers. Following exposure, all listeners in an experiment were tested on one or more *test conditions*, each containing a particular set of L2 speech recordings. Higher recognition accuracy during test indicates better generalization from exposure. The intelligibility of the different types of L2 speech in the three datasets ranges from $< 33\%$ (two of every three words misunderstood by L1-English listeners) to $> 90\%$.

Table 1 summarizes all exposure-test conditions contained in the data. The data span generalization across talkers within the same L2 accent (e.g., generalizing from

one Mandarin-accented English talker to another [6–8]) and across different L2 accents (e.g., generalizing from Spanish- to Korean-accented English [7, 8]), for both isolated word recordings [8] and connected sentence recordings [6, 7].

We first assessed the self-predictability of the data to avoid fitting to noise. One data sets had low self-predictability [7]. Results for these data are reported in the SI (??). The remaining two data sets contained XXX unique exposure-test combinations with a total of XXX word-level accuracy measures from a total of XXX participants (for details, SI ??).

Segment-, word-, and sentence-boundaries

All data sets contained transcriptions of all exposure and test recordings. We used the Montreal Forced Aligner [1] to obtain segment, word, and sentence boundaries by aligning the transcripts with the recordings. All boundaries were checked, and where necessary corrected, by the first author. The boundary annotations are available in the OSF repository for this project.

Transforming speech into latent representations

All speech recordings from the exposure and test phases of the three experiments were transformed into latent feature representations learned by a deep neural network model (DNN) trained for speech recognition (Fig 1). Specifically, we used the *HuBERT-large* model [17], which consists of 7 convolutional neural network (CNN) layers and 24 transformer (Tr) layers. The model is trained with self-supervised learning (SSL) on 60,000 hours of English. Model ablation MA1 supports identical conclusions for the HuBERT-large+ASR model, which is fine-tuned on 960 hours of labeled speech for the objective of automatic speech recognition (ASR).

While the feature representations learned will likely differ depending on the specific model architecture and training data, recent work found little difference in the extent to which alternative models (e.g., *Whisper* [23] or *wav2vec* [24]) can predict human speech perception [20, 21]. Model ablation MA2 confirms identical conclusions when *Whisper* or *wav2vec* are used instead of HuBERT.

Some steps in our procedure are compute-heavy. Most notably, the compute time for the temporal alignment of different segments, words, and sentences (see XXX) scales linearly with the number of latent features. We used *t*-SNE [18] to reduce the high-dimensional feature representations learned by HuBERT-Large from 1024 to 3 feature dimensions. Model ablations MA3.a-c suggest that this loss in information did not qualitatively affect conclusions.

For each combination of data set and DNN layer, we obtained separate *t*-SNE-ed 3D latent representations. Adaptive pooling was used to downsample the temporal resolution of all layers to 50 Hz (20ms intervals of speech) prior to applying *t*-SNE.¹ We used the segment-, word-, and sentence-boundaries to obtain segment-, word-, and

¹Prior to this step early CNN layers have higher temporal resolution (from 3200 Hz for CNN-1 to 50 Hz at CNN-6 and all transformer layers).

sentence-level feature representations for all exposure and test stimuli, separately for each combination of data set and DNN layer. These feature representations formed the input to all subsequent analyses.

Acoustic and auditory controls

We calculated to control measurements that were subjected to the same analyses as the DNN-based latent feature representations: 39-dimensional Mel-Frequency Cepstral Coefficients (MFCCs) and XXX-dimensional Spectro-Temporal Receptive Fields (STRFs).

MFCCs were used as the input to Hidden-Markov Model (HMM) ASR systems from the 80s to the early 00s [?]. They continue to be employed in the cognitive sciences as a lower-dimensional representation of the speech signal and its dynamics. Of the 39D MFCCs used here, the first 12 MFCCs represent the energy over 12 Mel-spaced frequency bands that are relevant to human speech perception. The 13th MFCC captures the overall energy over the speech signal. MFCC 14-26 and 27-39 summarize the first and second derivative, respectively, of the first 13 MFCCs. Unlike DNN-based latent representations, MFCCs are not the result of self-supervised learning, and thus are not shaped by the objective to predict contextual or missing data from the observed input.

The second measure is a XXX-dimensional summary of the auditory percept that is often used in auditory neurosciences []: spectro-temporal receptive fields (STRFs). STRFs convolve the acoustic input in a way that is thought to approximate the latent auditory representations in early auditory cortex. Each STRF is a 2D filter—over frequency and time—that models how a neuron or population of neurons responds to the stimulus. STRFs are not specific to speech, and are not shaped by the objective to predict contextual or missing data from the observed input.

Quantifying exposure variability

The hypothesis that high-variability exposure (HVE) supports generalization has not previously been formalized. We developed 16 different operationalizations of *exposure variability*. All 16 measures used the DNN-derived layer-specific latent representations—the ‘hyper-noodles’—of all exposure recordings in a particular exposure condition to quantify the exposure variability for that condition (Fig 3). The measures differed, however, in i) the level of linguistic organization at which they assess variability (segments, words, sentences), ii) whether they are defined with regards to types or tokens at that level of organization, iii) whether they capture variability between or within linguistic units at that level of organization, and iv) whether the measure is sensitive to the temporal structure of speech (all measures were sensitive to its spectral structure). Formal definitions of all measures are provided in the SI (??).

Quantifying exposure-test similarity

DTW – $\hat{c}$ listeners can adjust for speech rate difference within and across talker [REFS].

Model evaluation and cross-validation

Model ablations

MA1: Fine-tuning to task objective

MA2: DNN architecture

MA3: Dimensionality reduction

Compared to linear dimensionality reduction like PCA, *t*-SNE is computationally demanding. We chose *t*-SNE because it is known to preserve the local neighborhood structure—i.e., the relative placement of local tokens in the high-dimensional space []. This suits the present purpose since both the HVE and SBI hypotheses are formulated with reference to perceptual exemplars, for which locality has been argued to be particularly relevant []. MA3.b suggests that the choice of *t*-SNE was warranted, though the same qualitative conclusions are supported by UMAP dimensionality reduction, which focus on preserving global structure. MA3.c suggests that PCA indeed would not have been suitable for the current purpose as it loses too much information.

Similarity measure

Supplementary information. Supplementary information (SI) can be found on OSF at [URL/DOI].

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Declarations

Data Availability

The human behavioral data analyzed in this study were obtained from three previously published studies [6–8], and are publicly available on our Open Science Framework (OSF) repository at [URL/DOI]. The full-dimensional latent representations extracted from the different latent layers of pre-trained HuBERT model, as well as seeded examples of *t*-SNE-reduced transformations of these representations, are available in the same repository.

Code Availability

The computational framework combining the pre-trained HuBERT architecture and the exemplar-theoretic memory module was implemented in Python (version 3.X) using PyTorch (version 2.X) and the Hugging Face transformers library. The complete source code, including custom scripts for feature extraction, representation alignment, hypothesis formalization, and statistical evaluation against the human behavioral data, is openly available via GitHub at [Insert repo URL], also linked to our OSF repository at [URL/DOI]. A permanent, version-controlled repository containing the exact code

used to generate the figures and findings reported in this study has been archived on Zenodo under DOI [Insert DOI, e.g., doi.org].

Author Contributions

Writing: *initial draft*—TFJ, *revisions*—all authors; Computational models: *development & implementation*—ZJ with feedback from TFJ & YZ; Data analysis: *conceptualization*—TFJ & YZ, *implementation*—ZJ with feedback from TFJ & YZ, *interpretation*—all authors; Visualization: *conceptualization*—TFJ & ZJ, *implementation*—ZJ with feedback from TFJ & YZ.

Appendix A Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.

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Table 1: Exposure-test conditions of all datasets. We named conditions based on the type of generalization they tested. The condition names originally employed in the articles that published the data set are provided in the final column.

Condition		Exposure L1(s) (all L1/L2-English)	Test L1 (L2-English)	Unique combos	Data	Original name
control	(multi-talker)	—	$KOR^1 \times 6^2$	2	AN19	no training
		—	SPA $\times$ 6	2	AN19	no training
		—	BRP	1	B23	control
		—	FAR	1	B23	control
		—	SPA	1	B23	control
		—	TUR	1	B23	control
cross-accent	(single-accent, single-talker)	ENG $\times$ 5	MAN	4	X21	control
		FAR	BRP	1	B23	FAR
		SPA	BRP	1	B23	SPA
		TUR	BRP	1	B23	TUR
		BRP	FAR	1	B23	BRP
		SPA	FAR	1	B23	SPA
		TUR	FAR	1	B23	TUR
		BRP	SPA	1	B23	BRP
		FAR	SPA	1	B23	FAR
		TUR	SPA	1	B23	TUR
		BRP	TUR	1	B23	BRP
		FAR	TUR	1	B23	FAR
		SPA	TUR	1	B23	SPA
		KOR $\times$ 6	SPA $\times$ 6	2	AN19	different accent
		SPA $\times$ 6	KOR $\times$ 6	2	AN19	different accent
within-accent cross-talker	(multi-accent, multi-talker)	ALB, BEN, DUT, HIN, JAP, ROM	KOR $\times$ 6	1	AN19	mixed
		ALB, BEN, DUT, HIN, JAP, ROM	SPA $\times$ 6	1	AN19	mixed
		FRE, GER, SOM, RUS, MAN, TUR	KOR $\times$ 6	1	AN19	mixed
		FRE, GER, SOM, RUS, MAN, TUR	SPA $\times$ 6	1	AN19	mixed
		FAR, SPA, TUR	BRP	1	B23	noBRP
		BRP, SPA, TUR	FAR	1	B23	noFAR
		BRP, FAR, TUR	SPA	1	B23	noSPA
		BRP, FAR, SPA	TUR	1	B23	noTUR
		MAN	MAN	20	X21	single-talker
		KOR $\times$ 6	SPA $\times$ 6	2	AN19	same accent
		SPA $\times$ 6	KOR $\times$ 6	2	AN19	same accent
		MAN $\times$ 5	MAN	4	X21	multi-talker
within-talker	(multi-talker)	FAR, SPA, TUR	FAR	1	B23	noBRP
		FAR, SPA, TUR	SPA	1	B23	noBRP
		FAR, SPA, TUR	TUR	1	B23	noBRP
		BRP, SPA, TUR	BRP	1	B23	noFAR
		BRP, SPA, TUR	SPA	1	B23	noFAR
		BRP, SPA, TUR	TUR	1	B23	noFAR
		BRP, FAR, TUR	BRP	1	B23	noSPA
		BRP, FAR, TUR	FAR	1	B23	noSPA
		BRP, FAR, TUR	TUR	1	B23	noSPA
		BRP, FAR, SPA	BRP	1	B23	noTUR
		BRP, FAR, SPA	FAR	1	B23	noTUR
		BRP, FAR, SPA	SPA	1	B23	noTUR
		BRP	BRP	1	B23	BRP
		FAR	FAR	1	B23	FAR
		SPA	SPA	1	B23	SPA
within-talker	(single-talker)	TUR	TUR	1	B23	TUR
		MAN	MAN	4	X21	talker-specific

¹Language key: **ALB**anian, **BEN**gali, **BRP**-Brazilian Portuguese, **DUT**ch, **ENG**lish (United States), **FAR**si, **FRE**nch, **GER**man, **HIN**di, **JAP**anese, **KOR**ean, **MAN**darin Chinese, **ROM**anian, **RUS**sian, **SPAN**ish (Mexico), **SOM**ali, **TUR**kish

²The number of talkers per accent (if ~ 1)